

- 4 Element Z burns in nitrogen to form a solid, Z_xN_y . The monatomic nitride anion in Z_xN_y has a full outer shell of electrons.

Which pair of statements about this reaction is correct?

- A Each nitrogen atom gains three electrons. The formula of the solid is ZN_3 .
- B Each nitrogen atom gains three electrons. The formula of the solid is Z_3N .
- C The nitride ion has a $3+$ charge. The formula of the solid is ZN_3 .
- D The nitride ion has a $3+$ charge. The formula of the solid is Z_3N .